

A Simulation on Disease Transmission and the effects of Immunization

Project Topic 3 (Pandemic Flu Spread)

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Abstract

This project examines how a disease outbreak spreads across a population through a simplified classroom example with 51 students. By simulating daily contact over the course of the illness, we track how the virus moves through the group and determine the typical duration of an epidemic. The study further investigates the impact of a 60% immunization rate to observe how the immunity affects the outbreak's severity. Our results demonstrate that immunization significantly limits the reach of the virus, reducing the expected number of cumulative infections by almost 8 times, illustrating how herd immunity can effectively contain a widespread infection.

Problem Statement

This project investigates the spread of a disease within a small population of 51 elementary school students. The main problem is: how does a single infectious individual trigger a chain of infections, and how long does such an epidemic last?

On Day 1, patient zero (Tommy) arrives infectious while the remaining 50 students are healthy and fully susceptible. Tommy remains infectious for 3 days, and on each of those days, he independently infects any given susceptible student with probability $p = 0.01$. Any student infected by Tommy becomes infectious the following day for a 3-day period, and spreads the flu under the same transmission probability, resulting in a cascade of subsequent infections.

The problem is designed to be answered using both statistical and simulation-based approaches. We first determine the distribution of the infection and the expected number of infections Tommy causes on Day 1 and 2 using statistical approaches. Using simulation methods, we estimate the expected number of infected students each day for the duration of the epidemic, as well as a second scenario where each student has a 60% chance of being immunized, examining how herd immunity affects the size and duration of the outbreak.

Findings

These findings are similar to a real-life disease outbreak situation with a susceptible, infectious, and recovered (SIR) model. While the findings are over simplified for this classroom setting, it still models a rough view of how the SIR process develops over time.

a) Distribution of Infections on Day 1

On Day 1, Tommy is the only infectious person and there are 50 susceptible kids. For each kid, there is a 1% chance of being infected. Since each infection is independent and identical and each infection is a Bernoulli trial, the distribution of infections follows a **binomial distribution**.

Therefore, the distribution is: **Binomial(n=50, p=0.01)**

b) Expected number of Infections on Day 1

To calculate the expected number of infections for a Binomial distribution, we use the formula $E[X] = n * p$.

Therefore, the expected number of infections on day 1 = $E[X] = 50 * 0.01 = \mathbf{0.5 \text{ kids}}$

c) Expected number of Infections by Day 2

To calculate the expected number of infections by day 2, we will need to account for both the infections from Tommy on day 2 and the secondary infections by the kids who were infected on day 1, since infected children remain infectious for 3 days. We will also need to account for the reduction in the susceptible kids population as the number of infected children increases.

By day 2, there are 3 groups of infected kids, infected kids from day 0, day 1, and day 2. These 3 groups will be summed up to determine the expected number of infections at the end of day 2.

Day 0: Initial patient zero (Tommy) = **1 kid**

Day 1: Infected kids on day 1 (from part b) = $50 * 0.01 = \mathbf{0.5 \text{ kids}}$

For infections on day 2, we will need to recalculate the susceptible population and infection probabilities to account for the day 1 infected kids before evaluating the infections from day 2.

Susceptible population: $50 - 0.5 = \mathbf{49.5}$

- Susceptible population is reduced by 0.5 since 0.5 kids were infected on day 1

Infection probability: $1 - (0.99)^{1.5} = 0.015 \text{ (rounded)}$

- Joint infection probability from both Tommy and the newly infected kids from day 1

With the new susceptible population and infection probability for day 2, we can now calculate the expected number of infected kids on day 2.

Day 2: Infected kids on day 2 = $49.5 * 0.015 = \mathbf{0.7425 \text{ kids}}$

Therefore, the total expected number of infections by day 2 = Day 0 + Day 1 + Day 2
 $= 1 + 0.5 + 0.7425 = \mathbf{2.2425 \text{ kids}}$

d) Simulation of Infections in the classroom (0% Immunity)

The simulation was conducted with 50,000 trials to reduce statistical errors. A primary assumption of this model is that once a child recovers from the virus, they acquire total immunity and are removed from the susceptible pool for the remainder of the trial.

In the initial phase of the epidemic, the infection count increases rapidly as the virus transitions from Tommy to subsequent generations. In the growth phase, the expected cumulative count reaches about 15 students by day 14. As the pool of susceptible individuals is depleted, the growth rate tapers off. The epidemic typically stabilizes at a cumulative average of 18 infected students (Figure 1).

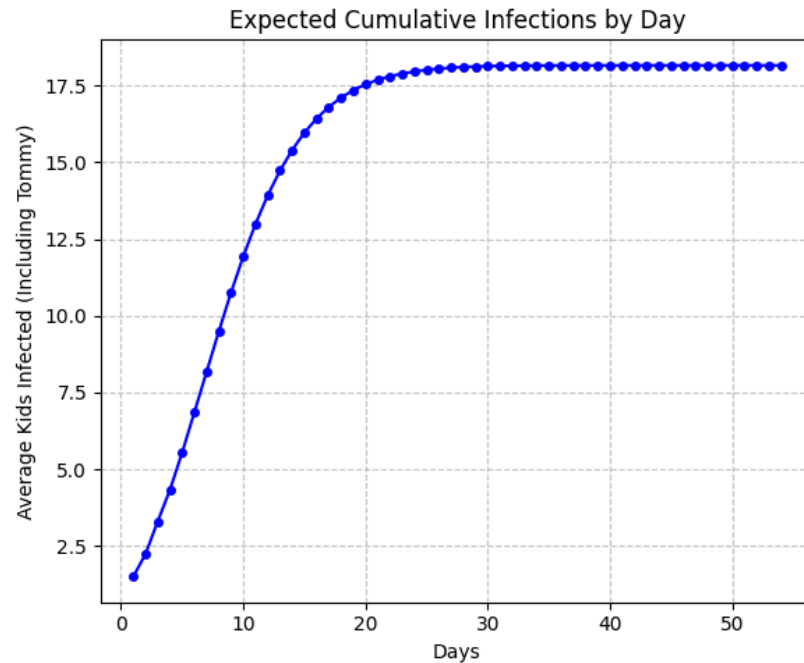


Figure 1: Expected Cumulative Infections by Day (0% Immunity)

The duration of the epidemic is defined as the time from when Tommy is infected to the recovery of the last infectious individual. The descriptive statistics for the duration are as follows:

Metric	Value
Mean Duration	13.7 Days
Median Duration	15 Days
Maximum Duration	49 Days

Table 1: Descriptive Statistics of Epidemic Duration (0% Immunity)

The duration histogram (Figure 2) reveals three distinct peaks, each representing a specific epidemiological scenario. This concept of branching is the Galton-Watson Process, where each branch (or infection) is exactly identical to the other branches, such that it creates stepped peaks in the early phases of the epidemic when the initial spread happens over a fixed window (Kimmel & Axelrod, 2002).

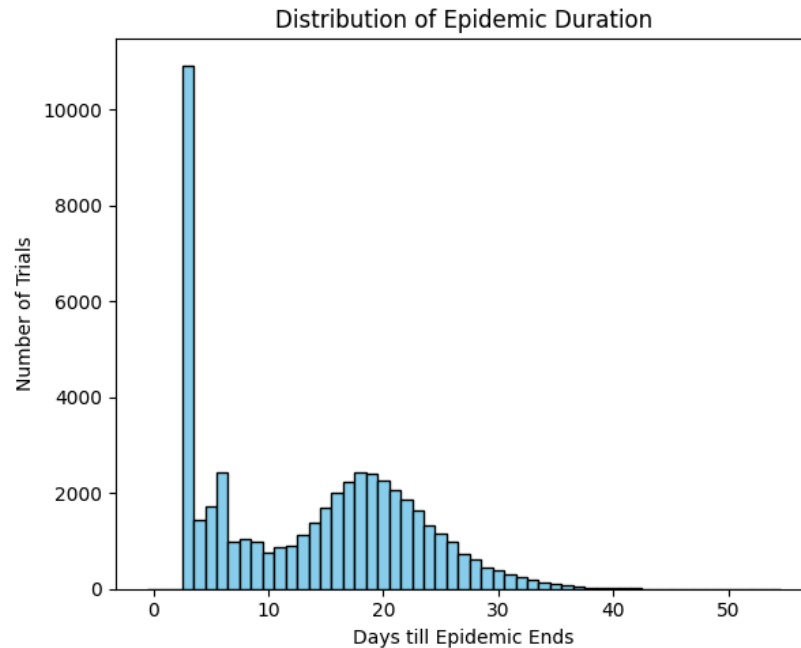


Figure 2: Epidemic Duration Histogram (0% Immunity)

1. **Primary Peak (Day 3):** Approximately 20% of trials conclude on Day 3. This represents the case where Tommy recovers without successfully transmitting the virus to any susceptible kids.
2. **Secondary Peak (Days 4–6):** Accounting for roughly 10% of trials, this peak represents a limited transmission scenario. In these cases, Tommy successfully infects a second generation of students. However, this second generation fails to further transmit the virus. The epidemic ends when this second generation recovers between days 4 and 6, depending on when they were infected.
3. **Tertiary Peak (Days 18–19):** Approximately 10% of trials result in a full outbreak. In this scenario, the virus achieves sustained transmission through multiple generations, infecting a significant portion of the classroom. The cluster around Days 18 and 19 reflects the typical timeframe for the virus to run out of susceptible hosts and for all infected individuals to fully recover.

Cases extending beyond Day 19 (up to the maximum of 49 days) represent more unlikely scenarios where the virus persists through a slow, linear chain of transmission rather than a rapid exponential spread.

e) Simulation of Infections in the classroom (60% Immunity)

In this iteration, a 60% immunization probability was applied to each student. The simulation initializes by determining the immunity of each kid through independent Bernoulli trials. Assuming that the immunization provides complete protection, children who are immunized are removed from the susceptible pool entirely. Following this initialization, 50,000 trials were executed to observe the impact of immunization.

The introduction of immunization significantly diminishes the spread of infections (Figure 3). While the initial trend shows an increase due to infections from Tommy, the transmission quickly tapers off within the population. Similar to the non-immunized scenario,

there is a marginal ramp-up in infections in the first few days. Due to the reduced susceptible population, the cumulative infections tapers off significantly earlier by Day 10. The expected cumulative number of infections stabilizes at about 2.23 students, representing a more controlled environment where the virus fails to spread exponentially.

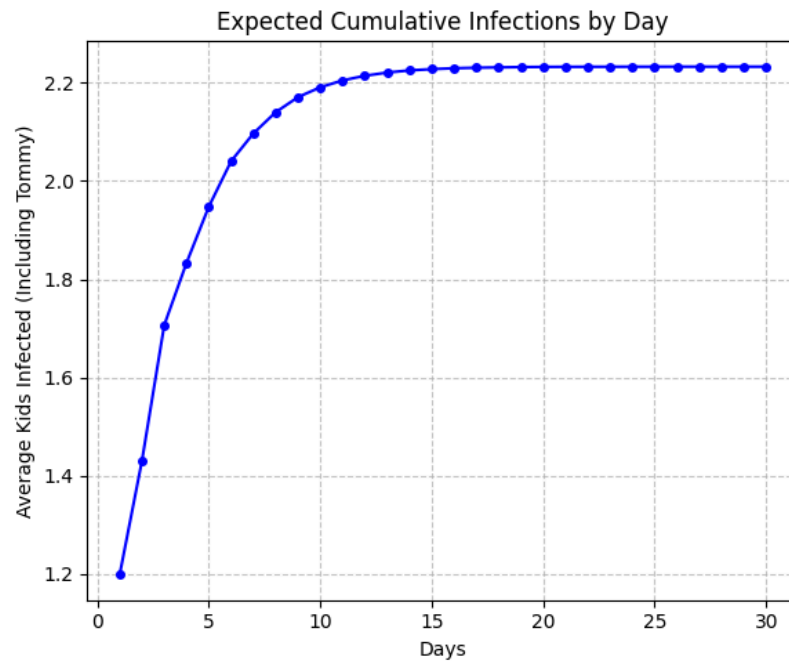


Figure 3: Expected Cumulative Infections by Day (0% Immunity)

With immunization, the duration of the epidemic is drastically reduced. The smaller susceptible population forces the virus transmissions to stop prematurely.

Metric	Value
Mean Duration	4.8 Days
Median Duration	3 Days
Maximum Duration	30 Days

Table 2: Descriptive Statistics of Epidemic Duration (60% Immunity)

The epidemic duration histogram for the immunized scenario highlights the lack of the long-term outbreak scenario, leaving only two primary peaks (Figure 4).

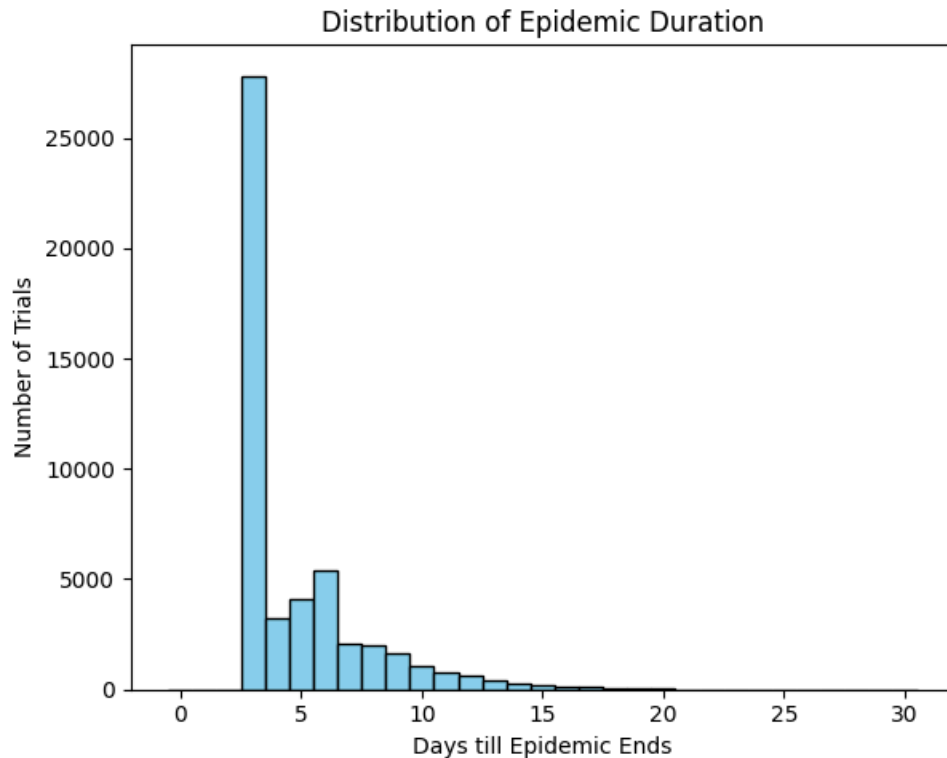


Figure 4: Epidemic Duration Histogram (60% Immunity)

1. **Primary Peak (Day 3):** This occurs in approximately 55% of trials. In more than half of the simulations, Tommy fails to transmit the virus to any susceptible kid, thereby ending the epidemic upon his recovery on Day 3.
2. **Secondary Peak (Days 4–6):** This represents roughly 25% of trials. In these instances, Tommy successfully infects a second generation of students but fails to propagate further. Combined with the first peak, this indicates that 80% of all epidemics are eradicated upon recovery of the second generation.

The most significant finding in the immunized model is the absence of the third peak, the full outbreak scenario previously observed around day 18 or 19. By reducing the susceptible population, the virus is unable to sustain a consistent chain of transmission beyond the immediate contacts of Tommy (Bullen & Jamrozik, 2023). This simulation demonstrates the efficacy of herd immunity, where if a sufficiently large portion of the population is protected, the probability of a widespread outbreak becomes significantly less likely, effectively curbing the spread of diseases within the community.

Conclusion

This simulation illustrates the containment of an infectious disease from a volatile threat to a controlled state through herd immunity. In a fully susceptible classroom of kids, outcomes are primarily split among immediate extinction, limited transmission, and full-scale outbreaks. By introducing a 60% immunization rate, it effectively eliminates the full-scale outbreak scenario, illustrating that high levels of immunization mitigates transmissions and protects the community.

However, the findings from this simplified model are not representative of reality. Without a stochastic distribution of infection and recovery times, the spread of the disease becomes much more predictable, resulting in a Galton-Watson process and the stepped peaks in the early infection stages. With a stochastic distribution, these stepped peaks will smooth out and the phases of the epidemic can be more accurately modelled.

In addition to stochastic infection and recovery times, more complex scenarios such as social network clusters, variation in population, and isolation can be built to model beyond the simplistic case of a homogenous population. Considering such cases will help to better understand how these scenarios play a part in an epidemic and further increases the model's fidelity. These enhancements would provide a more nuanced simulation to evaluate the effectiveness of public health strategies in real-world settings.

References

1. Bullen, M., Heriot, G. S., & Jamrozik, E. (2023). *Herd immunity, vaccination and moral obligation*. Retrieved April 19, 2026, from <https://pmc.ncbi.nlm.nih.gov/articles/PMC10511978/>
2. Kimmel, M., & Axelrod, D. E. (2002). *Branching processes in biology*. Retrieved April 21, 2026, from https://www.researchgate.net/profile/Marek-Kimmel/publication/228792062_Branching_Processes_in_Biology/links/54e420bf0cf282dbed6e9876/Branching-Processes-in-Biology.pdf